

BPG-16 · BEST PRACTICE GUIDES

Risk registers that work

Turning a compliance artefact back into a management instrument

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

Most risk registers are maintained and not used. This guide sets out what separates the two: risk statements built as cause, event and effect; why an effect such as "delay to project" cannot be managed as a risk; what qualitative probability-impact scoring can and cannot order; ownership and action dates as the register's live parts; the link between the register and contingency; and the health metrics that tell you, in a few minutes, whether your register is a management tool or wallpaper. The worked example shows two risks with identical matrix scores whose expected values differ by a factor of more than three.

— About this document

Document	BPG-16
Series	Best Practice Guides
Version	1.0
Status	draft
Date	2026-08-04
Unresolved placeholders	0

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Risk registers that work

In one paragraph. Most risk registers are maintained and not used. This guide sets out what separates the two: risk statements built as cause, event and effect; why an effect such as "delay to project" cannot be managed as a risk; what qualitative probability-impact scoring can and cannot order; ownership and action dates as the register's live parts; the link between the register and contingency; and the health metrics that tell you, in a few minutes, whether your register is a management tool or wallpaper. The worked example shows two risks with identical matrix scores whose expected values differ by a factor of more than three.

Who this is for. Risk managers and project controls managers who own a register; cost engineers and planners who have to turn one into contingency or a schedule model; and project managers who chair the review that the register is supposed to serve.

— 1. The register that nobody uses

Almost every project has a risk register. A small minority have one that changes a decision in any given month. The gap is not effort — unused registers are often meticulously maintained — and it is not tooling. It is that the register was built to demonstrate that risk management is happening rather than to make risk decisions, and an artefact built for demonstration acquires a predictable shape: many entries, vague statements, team-level ownership, scores that never move, actions without dates, and a review that reads the top ten aloud and closes.

The diagnosis matters because the remedies differ. A register failing from neglect needs attention. A register failing from purpose needs rebuilding around the decisions it is meant to inform: what we fund, what we mitigate, what we escalate, what we tell the client.

— 2. Cause, event, effect — and why the structure is not pedantry

A risk entry that can be managed has three separable parts.

The cause is a condition or dependency that exists now. It is not uncertain; it is a fact about the project. "The permanent power connection is delivered by a utility outside the contract."

The event is the uncertain thing. It either happens or it does not, and a probability attaches to it. "Energisation is granted later than the commissioning window requires."

The effect is what the event does to an objective, measurable in time, money, scope or quality. "Systems commissioning is delayed and time-related site costs extend."

Written as a single entry: because the permanent power connection depends on a utility outside the contract, there is a risk that energisation is granted later than the commissioning window requires, which would delay systems commissioning and extend time-related site costs.

The structure earns its place because each part drives a different management act. The **cause** is what a mitigation attacks — you cannot reduce an event's probability except by acting on its cause,

which is why registers full of causeless risks generate mitigations that are really just monitoring. The **event** carries the probability and can be declared closed when the uncertainty resolves. The **effect** carries the impact, feeds quantification, and determines who should own the entry, because a risk belongs to whoever owns the objective it threatens, not to whoever first mentioned it.

Two tests. If the cause is not a present fact it is a second risk, and the two should be separated. If the event cannot be declared to have happened or not happened on a given date it is a condition, and it belongs in the assumptions or the issue log.

— 3. Why "delay to project" is not a risk

"Delay to project" is an effect. So is "cost overrun", "reputational damage" and "loss of client confidence". A register of effects cannot be managed, for three reasons that compound.

No action addresses it. Delay arises from dozens of unrelated causes with different owners. An entry aggregating them can only be assigned to the project manager, and the mitigation can only be "manage the project". That is a job description, not a control.

It can never be closed. The project is at risk of delay from first day to last, so the entry sits permanently with its score unchanged — teaching every reader that entries on this register do not move. One immortal entry is enough to establish the culture.

It double-counts. If the register also carries the specific events that would cause delay, the aggregate entry counts them again. Any quantification built on it is then wrong by an unknown amount, in the direction of overstating exposure — which sounds prudent until a contingency request must be defended line by line.

The same logic disposes of two other common entries. "Poor contractor performance" is an assessment, not an event; the events are the specific things a poorly performing contractor does — misses a delivery date, fails an inspection, under-resources a work front — each with its own cause and response. And something that has already happened is not a risk: it is an issue, and it belongs in the issue log with an action and a date. A register carrying issues has stopped distinguishing what might happen from what has, which is the one distinction it exists to make.

— 4. Qualitative scoring: what it orders, and what it cannot

Almost every register scores probability and impact on ordinal bands — typically one to five — and multiplies them into a matrix score. Used for what it is good at this is sound and fast: it triages a large register quickly, needs no cost model, works in a workshop with people who will not tolerate a quantitative exercise, and identifies which entries deserve quantification.

Its limitation is structural, not a matter of calibration. **The bands are ordinal and the impact scale is usually geometric, while the score is arithmetic.** Band 4 is not twice band 2 in money; it is often ten or twenty times. Multiplying two rank positions therefore produces a number with no monetary meaning, and two entries in the same cell can differ in expected value by an order of magnitude. Section 8 works this through.

Three consequences for practice.

Use the matrix to triage, not to rank for funding. It answers "which entries deserve more analysis", not "which entries drive the contingency". Any register whose top ten is selected by matrix score has the wrong ten in it.

Define the bands in the units of the project, and write them on the register. "Low, medium, high" is re-interpreted by everyone who uses it; a stated money range and a stated probability range are at least argued about consistently. Calibrate so that band 5 is genuinely material — a band-5 threshold set at a figure the project spends monthly makes the whole top row meaningless.

Score impact against a stated objective. A risk can be band 2 for cost and band 5 for schedule. One combined score hides which objective is threatened, and therefore who should own it.

Where a project has an appetite statement, the matrix is also where it is expressed — by drawing the escalation line on the grid rather than leaving each chair to apply their own tolerance. That is professional judgement made explicit, which is the only form in which it can be reviewed.

— 5. Ownership, actions and dates — the live parts

The columns that determine whether a register is alive are not the scoring columns.

Owner must be an individual. "Engineering" cannot receive a phone call. Where a risk genuinely spans functions, name one owner and list the others as contributors; a shared owner is an unowned risk.

Every response has an action, an owner and a date in the future. An action without a date is an intention. A register where a third of the action dates have passed has told you how much attention it receives, more reliably than any maturity assessment.

Responses are typed, and the type is honest. The threat strategies are avoid, transfer, reduce and accept; the opportunity strategies are exploit, share, enhance and accept. Most entries claiming to be mitigated are in fact accepted with monitoring, and calling that "mitigate" inflates the project's apparent control. Conscious acceptance with contingency held against it is legitimate and often correct; disguising acceptance as mitigation is not.

A response is justified when it reduces probability times impact by more than it costs. That test is arithmetic and should be shown: the register carries the pre-response and post-response assessment and the response's cost, so the decision is visible. Responses whose cost exceeds the exposure they remove are recorded as rejected with a reason, which is what stops the same proposal returning every quarter.

Every entry carries a review date. A risk whose score has not moved in three cycles is either genuinely stable — in which case say so and review it less often — or unexamined. Recording which takes one column.

— 6. From register to contingency, and to the schedule model

Summing the **expected monetary value (EMV)** of every entry — probability multiplied by impact, added across the register — gives a first-pass figure for the contingency needed against identified risks. That sum is valid whatever the correlations between entries, because the expected value of a sum is the sum of the expected values.

It is also not the answer, for two reasons that pull in the same direction.

The expected value is not a fundable number. No individual risk costs its expected value; each costs either its impact or nothing. What a project funds is a confidence level on the distribution of total outcomes, and that level sits above the expected-value sum whenever risks can coincide.

Correlation fattens the tail without moving the mean. Real registers are full of shared drivers: one weather season behind several delays, one overheated supply market behind several vendor slips. They make coincidence far more likely than independent arithmetic suggests, raising the funded confidence level while leaving the expected-value sum exactly where it was.

The practical discipline needs no mathematics. For each pair of significant entries, ask what would make both happen. A nameable shared cause is a correlation to declare — to the modeller if there is a model, and to the contingency basis if there is not. [BPG-10 – Contingency and management reserve](#) owns how the funded level is chosen and governed; [BPG-17 – Quantitative schedule risk analysis](#) owns what happens when the register is mapped onto the schedule and simulated.

One connecting discipline belongs to the register rather than the model: **risk loading**. Each entry that will be quantified names the specific activities and cost lines it would strike. Without that mapping the model runs on a parallel set of assumptions and the register merely resembles it; with it, the register is the model's input and a change to one is visible in the other.

— 7. How this goes wrong

The register is a deliverable, not a tool. Produced because the framework requires one, reviewed because the review is in the schedule, consulted by nobody in between. The tell is that nothing in the project plan changes as a result. A register that has never caused a schedule change, a contingency draw or a contract negotiation is not managing risk.

Entries are effects, so nothing can be closed. Covered in §3. The symptom is a register whose population only ever grows.

The top ten is chosen by matrix score. The heavy-tail entries — low probability, very high impact — sit below the line because the matrix rewards likelihood. Those are usually the entries that determine the contingency and, occasionally, the ones that end the project.

Mitigations are monitoring in disguise. "Monitor closely", "maintain dialogue with the supplier", "escalate if required". None changes probability or impact, so the residual assessment should be unchanged — yet the register almost always shows a reduced residual score once they are added.

Everything is owned by the project manager. A register with one owner is a to-do list belonging to someone who already has one.

Impact is scored on the project, not on the objective. "High" means high for schedule in one entry and high for reputation in another, and the two are then compared as though commensurable.

Opportunities are absent. A register with no upside entries is evidence about the workshop, not the project, and it biases the contingency conversation, because what would net against threats is not on the page.

The register outlives its assumptions. Scores set at sanction are still there at handover. The risk profile a project starts with is not the one it carries, so re-identification at each gate is not optional.

Quantification double-counts responses. The entry is quantified at its pre-response impact and the response's cost is also in the budget. The exposure is funded twice and the request cannot survive scrutiny.

8. Worked example

Illustrative figures. Currency is USD, whole dollars, no escalation and no discounting applied. Impacts are cost impacts on a single objective. Probabilities are point estimates from a workshop.

8.1 The scales in use

Band	Probability	Cost impact
1	below 10 %	below 25,000
2	10 % to 25 %	25,000 to 250,000
3	25 % to 50 %	250,000 to 750,000
4	50 % to 75 %	750,000 to 2,500,000
5	above 75 %	above 2,500,000

Matrix score is the product of the two band numbers, on a scale of 1 to 25. The project's escalation line is drawn at a score of 8.

8.2 Three entries

Ref	Event (abbreviated)	Probability	P band	Impact	I band	Matrix score
R-11	Late utility energisation	60 %	4	100,000	2	8
R-24	Single-source vendor fails to deliver the main package	20 %	2	1,000,000	4	8
R-31	Contaminated ground discovered on the east plot	5 %	1	1,200,000	4	4

R-11 and R-24 share a matrix score of 8 and sit on the escalation line. R-31 scores 4 and sits below it.

8.3 Expected value

EMV = probability × impact

R-11: $0.60 \times 100,000 = 60,000$

R-24: $0.20 \times 1,000,000 = 200,000$

R-31: $0.05 \times 1,200,000 = 60,000$

Total EMV = $60,000 + 200,000 + 60,000 = 320,000$

8.4 What the two orderings disagree about

Same score, different money. R-11 and R-24 both score 8. Their expected values are 60,000 and 200,000.

$$\text{ratio} = 200,000 \div 60,000 = 3.33 \text{ (to two decimal places)}$$

R-24 carries more than three times the expected exposure of R-11, and the matrix places them in the same cell. A register that funds by matrix position funds these two identically.

Different score, same money. R-11 scores 8 and R-31 scores 4 — R-11 is ranked at twice R-31's priority. Their expected values are identical at 60,000 each. The matrix has ordered by likelihood, and likelihood is not what the project pays.

Where the tail lives. R-31 can cost 1,200,000 in a single event — 3.75 times the register's entire expected-value sum ($1,200,000 \div 320,000 = 3.75$) — and the matrix places it below the escalation line. If the top ten were selected by matrix score, the project's largest single loss would not be in it.

8.5 Why this happens, in one line

The impact bands step by roughly an order of magnitude while the band numbers step by one. Multiplying band numbers multiplies rank positions; multiplying probability by impact multiplies a proportion by an amount of money. Only the second produces a quantity you can add up, compare or fund.

8.6 What follows for this register

Use the matrix to decide which entries go forward for quantification, with the forward criterion set on either dimension so a band-4 or band-5 impact goes forward regardless of probability. Order the funding conversation by expected value, and hold the heavy-tail entries separately because their expected value understates what one occurrence does. Then ask what R-24 and R-31 have in common: a shared driver — one contaminated-land condition that also delays the vendor's foundation works, say — is the pairing that moves the funded level.

The assumptions travel with the numbers: probabilities are workshop point estimates with no range around them; impacts are single figures, not distributions; nothing here nets opportunities against threats; and the expected-value sum is a starting point for contingency, not a contingency figure. [BPG-10](#) takes it on.

— 9. Register health in five minutes

Illustrative figures from a single register at one review date.

A register of **84 entries**, of which:

- **61** have a named individual as owner: $61 \div 84 = 72.6 \%$. Twenty-three entries are owned by a team, a function or nobody.
- **31** have a next action date that has already passed: $31 \div 84 = 36.9 \%$.
- **44** have not changed score in three or more review cycles: $44 \div 84 = 52.4 \%$.
- **2** were added and **0** closed in the last cycle.

Each figure is a question rather than a verdict, but together they are decisive. A register where more than half the entries have not moved in three cycles, more than a third of actions are overdue and nothing closed all cycle is not being used; it is being maintained. And 84 is itself a finding — a project board cannot manage 84 of anything, so either the register needs a working subset with a stated selection rule, or most entries should be closed, merged or moved to a watch list on a longer cycle.

Run these counts before the review meeting rather than after it. They take minutes, they are not disputable, and they change what the meeting is about.

— 10. Checklist

Every entry

- [] Cause is a present fact, event is a discrete uncertainty, effect is measurable in time or money.
- [] The event can be declared to have happened or not happened on a stated date.
- [] The entry is not an effect, an assessment or an issue.
- [] Owner is a named individual who owns the threatened objective.
- [] Probability and impact are assessed against published bands, in project units.
- [] Cost impact and schedule impact are scored separately.
- [] Response is typed honestly — avoid, transfer, reduce, accept, or the opportunity equivalents.
- [] Response has an action, an owner and a date in the future.
- [] Pre-response and post-response assessments are both recorded, with the response's cost.
- [] Where the entry will be quantified, the activities and cost lines it would strike are named.

Every review

- [] Overdue action count, unowned count and unchanged-score count computed before the meeting.
- [] Entries closed as well as entries added.
- [] Heavy-tail entries reviewed regardless of matrix position.
- [] Shared drivers between significant entries named and recorded.
- [] Opportunities reviewed on the same footing as threats.

Every gate

- [] Re-identification run, not just re-scoring, with the people doing the next phase's work.
- [] Assumptions behind the original scores tested against what is now known.
- [] The expected-value sum reconciled to the contingency held, with the difference explained.

A register that passes this list is smaller than the one it replaced and is read by people who are not paid to read it. That is the only durable evidence that risk management is happening: the register changed what the project did, and someone can name the month it happened.

— Related

- [BPG-10 – Contingency and management reserve](#) — how the quantified register becomes a funded number, and who may draw it
- [BPG-17 – Quantitative schedule risk analysis](#) — what happens when the register is mapped onto the schedule and simulated
- [BPG-19 – Project controls assurance and health checks](#) — how a reviewer tests whether the register is genuinely in use
- [TPL-10 – Risk register](#) — the column standard and definitions this guide assumes
- [AIG-06 – AI for risk identification and quantification](#) — where machine assistance helps, and what must stay with the professional

— Sources and standards

- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 12 — Risk Management for Project Controls, first authored draft, August 2026: the cause-event-effect standard, qualitative and quantitative analysis, response strategies, correlation and shared drivers.
- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 3 — Budgeting and Forecasting, first authored draft, August 2026: estimating bias and the treatment of uncertainty in a budget.
- PCI Canonical Facts ([docs/publication-framework/00-framework/CANONICAL-FACTS.md](#)), verified August 2026: naming, status and claims policy.

ISO 31000 is the international risk-management standard most often referenced in this area. This guide reproduces none of its text, structure or clause numbering; where its principles are relevant — that risk management is integrated into decisions rather than run alongside them, and that it addresses uncertainty in both directions — they are explained in the Institute's own words. No survey data, benchmark or industry average is cited, because none was verified for this guide.

— Status and version

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